Arithmetic and complexity

An explicit rational Valiant-rigid matrix family

Difficulty: extreme **Importance:** broadly interesting

Provenance: rational-field specialization of a source-stated explicit-rigidity target.

Status: no construction meeting this target found in screening on 2026-09-08.

For $A \in \mathbb{Q}^{n\times n}$ and an integer $0 \le r \le n$, define its rigidity over $\mathbb{Q}$ by

$$R_A^{\mathbb{Q}}(r) = \min\left\{|\{(i,j) : A_{ij} \neq B_{ij}\}| : B \in \mathbb{Q}^{n\times n},\ \operatorname{rank}_{\mathbb{Q}}(B) \le r\right\}.$$

Construct a constant $\delta > 0$ and a family $A_n \in \mathbb{Q}^{n\times n}$ such that a single deterministic algorithm, on input the unary encoding 1^n, outputs all rational entries of A_n in binary using polynomially many bit operations in n, and

$$R_{A_n}^{\mathbb{Q}}\left(\left\lfloor \frac{n}{\log\log n} \right\rfloor\right) \ge n^{1+\delta}$$

for every sufficiently large integer n. Here log denotes the natural logarithm. The output convention requires polynomial total bit length; an exponentially long algebraic description does not meet this requirement.

Rigidity measures how resistant a matrix is to rank reduction by sparse entry changes. An explicit family at this scale would supply hard instances for linear circuit computation. The source states the target over a field; this entry explicitly selects rational entries and rational perturbations, rather than asserting that all choices of field are equivalent.

References

1. J. Alman and J. Liang, *Low Rank Matrix Rigidity: Tight Lower Bounds and Hardness Amplification*, arXiv:2502.19580 (2025), Section 1 and Section 1.1: definition of rigidity, uniform polynomial-time explicitness, and the $n/\log\log n$ Valiant target. Paper.
2. L. Hambardzumyan, K. Myasnikov, A. Riazanov, M. Shirley, and A. Shraibman, *Spiky Rank and Its Applications to Rigidity and Circuits*, ICALP 2026, LIPIcs 374, article 106, Section 1.2.1: explicit rigidity remains insufficient for the relevant circuit lower bounds. Published paper.
3. L. G. Valiant, *Graph-theoretic arguments in low-level complexity*, MFCS 1977, pp. 162–176, the original rigidity connection to linear circuits. Article.
4. Z. Dvir and A. Liu, *Fourier and Circulant Matrices are Not Rigid*, arXiv:1902.07334 (2019), main non-rigidity results for prominent structured candidates. Paper.

Status check — 2026-09-08

checked 2502.19580v1 (2025-02-26), the July 2026 published paper, and searches "matrix rigidity explicit rational 2026 Valiant construction" and "Valiant rigidity solved". The 2025 bounds for logarithmic target rank and its conditional amplification results do not give the displayed higher-rank target. Known non-rigidity results prevent treating Walsh or Fourier matrices as established candidates. No qualifying polynomial-time rational construction or resolution was found. This status applies to the stated rational specialization.